Homework 8: Exercise 5.1 Solution

Date

Q5 let $u = (x_1, 0)$ & $v = (x_2, 0)$

① $u + v$ is in V.

$(x_1, 0) + (x_2, 0)$

$= (x_1 + x_2, 0)$

now let $x_3 = x_1 + x_2$

$= (x_3, 0) \in V$ (closed under addition)

② ku is in V

$k(x_1, 0) = (kx_1, 0)$

let $kx_1 = x_2$

$(x_2, 0) \in V$ (closed under multiplication)

③ $u + v = v + u$

$(x_1 + x_2, 0) = (x_2 + x_1, 0)$

$(x_1 + x_2, 0) = (x_1 + x_2, 0)$

④ let $w = (x_3, 0)$

$u + (v + w) = (u + v) + w$

$u + v = (x_1 + x_2, 0)$

$v + w = (x_2 + x_3, 0)$

Taking LHS $(u + v) + w$

$(x_1 + x_2, 0) + (x_3, 0)$

$(x_1 + x_2 + x_3, 0)$

Date

$(x_1, 0) + (x_2 + x_3, 0)$

$u + (v + w)$

(5) $u + 0 = u$

$(x_1, 0) + (0, 0) = (x_1 + 0, 0)$

$= (x_1, 0)$

(6) $u + (-u) = 0$

$u = (x_1, 0)$

$-u = (-x_1, 0)$

$(x_1, 0) + (-x_1, 0)$

$(x - x_1, 0)$

$(0, 0)$

(7) $c(u+v) = cu + cv$

$c(x_1 + x_2, 0)$

$(cx_1, 0) + (cx_2, 0)$

(8) $(c+d)u = cu + du$

$(c+d)(x_1, 0)$

$(cx_1, 0) + (dx_1, 0)$

$cu + du$

Scanned with CamScanner

Date

(9) $c(du) = cd(u)$

Taking LHS.

$c\,(dx_1, 0)$

$(cdx_1, 0)$

$cd\,(x_1, 0)$

(10) $1u = u$

$1(x_1, 0)$

$(x_1, 0)$

It is a vector space.

Q6) For the set of all real numbers of the form (x, y) where $x \geq 0$, with standard operations on $\mathbb{R}^2$.

→ Not a vector space as one axiom fails here.
↳ there exists an object (x_K, y_K) such that
$(x, y) + (x_K, y_K) = 0$
This can't be true as $x \in \mathbb{R}^+$ hence $x_K \in \mathbb{R}^-$
which does not exist in the vector space.

11. The set of all real-valued functions f defined everywhere on the real line and such that $f(1) = 0$, with the operations defined in Example 4.

$$(\mathbf{f} + \mathbf{g})(x) = f(x) + g(x) \quad \text{and} \quad (k\mathbf{f})(x) = kf(x)$$

→ Checking all the axioms:-

→ $f(x) \in F(-\infty, \infty)$ $\{$such a vector field is denoted like this, dw$\}$.

and $g(x) \in F(-\infty, \infty)$
then $(f+g)(x) \in F(-\infty, \infty)$ → This must be true as $(f+g)(x) = f(x) + g(x)$ and the sum of 2 real valued functions is a real valued function.

→ $(f+g)(x) = (g+f)(x)$

$f(x) + g(x) = g(x) + f(x)$

LHS = RHS. (commutative addition).

$\rightarrow (k+f)(x) = (f+k)(x) = f(x)$.

if $k(x) = 0 \ \forall x$

Then the axiom is True and we know that $k(x) \in F(-\infty, \infty)$ as it is also a real valued function.

$\rightarrow f(x) + (g+h)(x) = (f+g)(x) + h(x)$.

$f(x) + g(x) + h(x) = f(x) + g(x) + h(x)$ → True.

$\rightarrow \exists\, g(x) \in F(-\infty, \infty)$ such that $g(x) + f(x) = f(x) + g(x) = 0$

if $g(x) = -f(x)$ The axiom is True and we know that $-f(x) \in F(-\infty, \infty)$ as it is a real valued function.

$\rightarrow kf(x) \in F(-\infty, \infty)$ true as $kf(x)$ is just another real valued function given that k is a real scalar.

$\rightarrow k((f+g)(x)) = kf(x) + kg(x)$.

$k(f(x) + g(x)) = kf(x) + kg(x)$.

$kf(x) + kg(x) = kf(x) + kg(x)$. ✓

$\rightarrow (k+m) f(x) = kf(x) + mf(x)$.

$kf(x) + mf(x) = kf(x) + mf(x)$. ✓

Scanned with CamScanner

$kf(x) + mf(x) = kf(x) + mf(x)$. ✓

$\rightarrow k(mf(x)) = (km)f(x)$.

↳ True.

$\rightarrow 1 \cdot f(x) = f(x)$

1 Times any real valued function is the function itself.

$\therefore f(x) \in F(-\infty, \infty)$ as it qualifies through all axioms.

17. Show that the following sets with the given operations fail to be vector spaces by identifying all axioms that fail to hold.

(a) The set of all triples of real numbers with the standard vector addition but with scalar multiplication defined by $k(x, y, z) = (k^2x, k^2y, k^2z)$.

(b) The set of all triples of real numbers with addition defined by $(x, y, z) + (u, v, w) = (z + w, y + v, x + u)$ and standard scalar multiplication.

(c) The set of all 2×2 invertible matrices with the standard matrix addition and scalar multiplication.

Q17)(a) $(k+m)\vec{u} = k\vec{u} + m\vec{u}$ fails here

$\Rightarrow (k+m)(x, y, z) = ((k^2+2km+m^2)x, (k^2+2km+m^2)y, (k^2+2km+m^2)z)$

while

$k\vec{u} + m\vec{u} = ((k^2+m^2)x, (k^2+m^2)y, (k^2+m^2)z)$

Q17)(b) $u + (v+w) = (u+v) + w$.

$(a_1, b_1, c_1) + ((a_2, b_2, c_2) + (a_3, b_3, c_3))$

$(a_1, b_1, c_1) + (c_2+c_3, b_2+b_3, a_2+a_3)$

$\rightarrow (a_2+a_3+c_1, b_1+b_2+b_3, c_2+c_3+a_1)$

$\hookrightarrow (u+v)+w$.

$(c_1+c_2, b_1+b_2, a_1+a_2) + (a_3, b_3, c_3)$

$\hookrightarrow (a_1+a_2+c_3, b_1+b_2+b_3, c_1+c_2+a_3)$ not a vector space

Scanned with CamScanner

Date

Q19

(a) let $u = (x_1, y_1)$, $V = (x_2, y_2)$

$y = mx \rightarrow$ line passing through origin.

$u + v \in V$, $(x_1 + x_2, y_1 + y_2)$

$\downarrow x_3 \quad \downarrow y_3$

$y_1 + y_2 = m(x_1 + x_2)$

$y_3 = mx_3 \rightarrow$ proved

(a) $y = mx$

$ky = mkx$ let $ky = y_1$

$y_1 = mx_1$ $kx = x_2$

$ku \in V$ proved

(b) $ax + by + cz = d$

$a(x_1 + x_2) + b(y_1 + y_2) + c(z_1 + z_2) = d$

$u + v \in V$

$kax + kby + kcz = d$

$ku \in V$ closed under both

Question 06

The set of all pairs of real numbers of the form (x, y) , where $x \geq 0$, with the standard operations on R^2.
Solution: It does not satisfy the axiom 5 of vector space. See the previous solution.

Question 11

The set of all real-valued functions f defined everywhere on the real line and such that $f(1) = 0$, with the operations defined in Example 4.
Solution: It satisfies all axioms of vector space, one can consider $f(x) = x - 1$ which satisfies $f(1) = 0$. See the previous solution.

Question 26

It was shown in Exercise 14 above that the set of polynomials of degree 1 or less is a vector space under the operations stated in that exercise. Is the set of polynomials whose degree is exactly 1 a vector space under those operations? Explain your reasoning.
Solution: It does not contain zero vector hence it is not vector space, i,e. $x - x = 0$

Question 27

Consider the set whose only element is the *moon*. Is this set a vector space under the operations $moon + moon = moon$ and $k(moon) = moon$ for every real number k? Explain your reasoning.
Solution: Yes, *moon* has to be the zero element because $moon + moon = moon$ means that '*moon*' fulfills the definition of the additive identity.

Question 28

Do you think that it is possible to have a vector space with exactly two distinct vectors in it? Explain your reasoning
Solution: No, the fact that all scalar multiples are included means that it has infinite vectors.

Q29

- Ax 7
- Hypothesis
- Ax 5
- Ax 4

$\hookrightarrow K0 + K4 = K4$

$$0 + \underbrace{K4}_{Z} = \underbrace{K4}_{Z}$$

$$0 + Z = Z$$

- Ax 4
- Ax 4
- if $K = 1$, then Ax 10 else hypothesis.

Date

Q3d

- Ax 1
- Ax 5
- hypothesis
- hypothesis

(32) No, it is not possible.

Consider a vector space with two zero vectors $\vec{O}_1$ and $\vec{O}_2$ such that for a vector $\vec{u}$

$$\vec{u}+\vec{O}_1=\vec{u} \quad -(1)$$

$$\vec{u}+\vec{O}_2=\vec{u} \quad -(2)$$

(1) $\Rightarrow$ $O_1=\vec{u}-\vec{u}$

(2) $\Rightarrow$ $O_2=\vec{u}-\vec{u}$

Hence, $\vec{u}-\vec{u}=\vec{u}-\vec{u}$

$$\vec{O}_1=\vec{O}_2$$

(33) No, it is not possible.

Consider a vector space ~~with~~ containing a vector $\vec{u}$ and a zero vector $\vec{O}$.

Consider that $\vec{u}$ has two negatives $\overrightarrow{(-u)}_1$ and $\overrightarrow{(-u)}_2$ such that

$$\vec{u}+\overrightarrow{(-u)}_1=\vec{O} \quad -(1)$$

$$\vec{u}+\overrightarrow{(-u)}_2=\vec{O} \quad -(2)$$

(1) $\Rightarrow$ $(-u)_1=\vec{O}-\vec{u}$

(2) $\Rightarrow$ $\overrightarrow{(-u)}_2=\vec{O}-\vec{u}$

$\therefore$ $\vec{O}-\vec{u}=\vec{O}-u \Rightarrow \overrightarrow{(-u)}_1=\overrightarrow{(-u)}_2$

Homework 8: Exercise Set 5.2 Solution

Question 01

Use Theorem 5.2.1 to determine which of the following are subspaces of R^3.
(a) All vectors of the form $(a, 0, 0)$.
Solution: Using Theorem 5.2.1
Let $u = (a, 0, 0)$ and $v = (b, 0, 0)$

$$u + v = (a, 0, 0) + (b, 0, 0) = (a + b, 0, 0) = (c, 0, 0)$$

$$ku = k(a, 0, 0) = (ka, 0, 0) = (d, 0, 0)$$

Hence, it is a vector space.
(b) All vectors of the form $(a, 1, 1)$.
Solution: Using Theorem 5.2.1
$u = (a, 1, 1)$ and $v = (b, 1, 1)$

$$u + v = (a, 1, 1) + (b, 1, 1) = (a + b, 2, 2) = (c, 2, 2)$$

Hence, it is not a vector space.
(c) All vectors of the form (a, b, c) , where $b = a + c$.
Solution: Using Theorem 5.2.1
Let $u = (a_1, b_1, c_1)$ and $v = (a_2, b_2, c_2)$

$$u + v = (a_1, b_1, c_1) + (a_2, b_2, c_2) \Rightarrow (b_1 + b_2) = (a_1 + a_2) + (c_1 + c_2) \Rightarrow b_3 = a_3 + c_3$$

$$ku = k(a, b, c) \Rightarrow k(b = a + c) \Rightarrow kb = ka + kc$$

Hence, it is a vector space.
(d) All vectors of the form (a, b, c) , where $b = a + c + 1$.
Solution: Using Theorem 5.2.1
Let $u = (a_1, b_1, c_1)$ and $v = (a_2, b_2, c_2)$

$$u+v = (a_1, b_1, c_1)+(a_2, b_2, c_2) \Rightarrow (b_1+b_2) = (a_1+a_2)+(c_1+c_2)+(1+1) \Rightarrow b_3 = a_3+c_3+2$$

Hence, it is not a vector space.
(e) All vectors of the form $(a, b, 0)$.
Solution: Using Theorem 5.2.1
Let $u = (a_1, b_1, 0)$ and $v = (a_2, b_2, 0)$

$$u + v = (a_1, b_1, 0) + (a_2, b_2, 0) = (a_1 + a_2, b_1 + b_2, 0) = (a_3, b_3, 0)$$

$$ku = k(a_1, b_1, 0) = (ka_1, kb_1, 0) = (a_4, b_4, 0)$$

Hence, it is a vector space.

Question 02

2. Use Theorem 5.2.1 to determine which of the following are subspaces of $M_{2\times 2}$
(a) All 2×2 matrices with integer entries
Solution: To be a subspace it needs to be closed under scalar multiplication (and addition); and the set of 2×2 matrices with integer entries is not. For example, take any 2×2 matrix with non-zero integer entries and multiply it by $1/2$; the resulting matrix will not be a matrix with integer entries and so it will not be in our set.
Mathematically,

$$\frac{1}{2}u = \frac{1}{2}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 1 \\ \frac{3}{2} & 2 \end{bmatrix}$$

(b) All matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a+b+c+d=0$

Solution: Let $u = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $v = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$ where $a+b+c+d=0$ and $a_1+b_1+c_1+d_1=0$, so by using Theorem 5.2.1

$$u+v = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} = \begin{bmatrix} a+a_1 & b+b_1 \\ c+c_1 & d+d_1 \end{bmatrix} = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$$

where $(a+a_1)+(b+b_1)+(c+c_1)+(d+d_1) = (a+b+c+d)+(a_1+b_1+c_1+d_1) = 0+0 = 0$

$$ku = k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

where $(ka)+(kb)+(kc)+(kd) = k(a+b+c+d) = k(0) = 0$. Hence it is a subspace.
(c) All 2×2 matrices such that $det(A) = 0$. It is a subspace.

Solution: Let $u = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $v = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, hence $det(u) = det(v) = 0$ but $det(u+v) = 1$. Hence it is not a subspace.

(d) all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

Solution: Let $u = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and $v = \begin{bmatrix} a_1 & b_1 \\ 0 & d_1 \end{bmatrix}$. It is a subspace, it is closed under addition and scalar multiplication , $u+v$ and ku.

$$u+v = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} + \begin{bmatrix} a_1 & b_1 \\ 0 & d_1 \end{bmatrix} = \begin{bmatrix} a+a_1 & b+b_1 \\ 0 & d+d_1 \end{bmatrix} = \begin{bmatrix} a_2 & b_2 \\ 0 & d_2 \end{bmatrix}$$

$$ku = k\begin{bmatrix} a & b \\ 0 & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ 0 & kd \end{bmatrix}$$

(e) all matrices of the form $\begin{bmatrix} a & a \\ -a & -a \end{bmatrix}$

Solution: It is not a subspace, it is not closed under scalar multiplication when k is negative.

Question 05

Use Theorem 5.2.1 to determine which of the following are subspaces of M_{nn}.
(a) all $n \times n$ matrices A such that $tr(A) = 0$
Solution: It is a subspace, one can show the following conditions.
Trace of a zero matrix is zero, so zero matrix must belongs here.

Any two matrices A_1, A_2, it is closed under addition. $A_1 + A_2 \Rightarrow tr(A_1 + A_2) = tr(A_1) + tr(A_2) = 0 + 0 = 0$. Addition is matrices is component-wise.
Scalar multiplication is also closed since $kA_1 \Rightarrow tr(kA_1) = 0$, $k(tr(A_1)) = k0 = 0$
(b) all $n \times n$ matrices A such that $A^T = -A$
Solution:

We will prove that it is a subspace. The zero vector 0 is the space, and it is skew-symmetric because $\mathbf{0}^T = \mathbf{0} = -\mathbf{0}$ Thus it is not empty set.

For condition, take arbitrary elements A, B. The matrices A, B are skew-symmetric, namely, we have $A^T = -A$ and $B^T = -B$. We show that $A + B$ belongs here. , or equivalently we show that the matrix A+B is skew-symmetric.

We have

$$(A + B)^T = A^T + B^T = -A + (-B) = -(A + B)$$

. Therefore the matrix A+B is skew-symmetric and condition 2 is met.

To prove the last condition, We show that kA is skew-symmetric.

$$(kA)^T = kA^T = k(-A) = -kA$$

. Hence kA is skew-symmetric.
(c) All $n \times n$ matrices A such that the linear system $Ax = 0$ has only the trivial solution.
Solution: It is not a subspace.
Since $Ax = 0$ has only trivial solution implies A^{-1} exists.
Let $u = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $v = \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix}$ are both invert-able but $u + v$ is not invert-able.
(d) All $n \times n$ matrices A such that $BA = AB$ for a fixed $n \times n$ matrix B
Solution: It is a subspace.
Zero matrix belongs to the subspace.

$$B\mathbf{0} = \mathbf{0}B = \mathbf{0}$$

Closed under addition

$$BA_1 + BA_2 = A_1B + A_2B \Rightarrow B(A_1 + A_2) = (A_1 + A_2)B \Rightarrow BA_3$$

Closed under scalar multiplication.

$$B(kA_1) = (kA_1)B = k(AB)$$